DESIGN AND IMPACT ANALYSIS OF HYBRID FOAM-FILLED VORONOI SANDWICH PANELS FOR VEHICLE CRUMPLE ZONES

ME3811 - PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

This project investigates the design and impact performance of a hybrid foam-filled Voronoi sandwich panel for application in automotive crumple zones. The proposed structure aims to overcome the limitations of conventional sandwich panels with uniform honeycomb cores, which often exhibit high peak impact forces and non-uniform collapse behaviour under dynamic loading conditions. In this work, Voronoi cellular geometries inspired by natural structures were developed using Rhino software, and a systematic design optimization was carried out using the Taguchi method. Nine different Voronoi configurations were evaluated through static structural analysis to identify the optimal cellular layout. Based on the optimized design, suitable materials were selected for the Voronoi sheets and the foam core, and a hybrid multi-layer sandwich panel was constructed. The crash performance of the developed structure was analysed using explicit dynamic finite element simulations under axial loading conditions representative of vehicle crumple zones. The results demonstrate that the foam-filled Voronoi sandwich panel exhibits improved progressive collapse behaviour, reduced peak impact force, and enhanced energy absorption capacity compared to conventional uniform-core sandwich panels. These findings indicate that hybrid Voronoi–foam sandwich structures have strong potential as efficient energy-absorbing components in automotive passive safety systems.

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LIST OF SYMBOLS

E	*Young's Modulus*
ρ	Density of material
σ	Stress
ε	Strain
F	Force
F_{max}	Peak Crushing Force (PCF)
F_{mean}	Mean Crushing Force (MCF)
v	Impact velocity
t	Thickness of Voronoi strut
T	Total thickness of panel
EA	Energy Absorption
SEA	Specific Energy Absorption
CFE	Crushing Force Efficiency
d	Deformation (displacement)
m	Mass of impactor
A	Cross-sectional area
v	Poisson's Ratio
K	Bulk Modulus
G	Shear Modulus
σ_γ	Yield Strength
E_t	Tangent Modulus

CHAPTER 1

INTRODUCTION

In recent years, the automotive industry has placed growing emphasis on enhancing vehicle safety while also reducing overall structural weight in pursuit of better fuel economy and lower emissions. Manufacturers and researchers are increasingly challenged to deliver designs that protect occupants more effectively without relying on heavier structures that can undermine efficiency targets. Within this wider safety agenda, crashworthiness remains a central performance requirement. Crashworthiness describes a structure's ability to manage and absorb impact energy during a collision, limiting the severity of deceleration experienced by occupants and helping to preserve the integrity of the passenger compartment.

A key contributor to crashworthiness is the crumple zone, most commonly located at the front and rear of a vehicle. Its role is not simply to "crush", but to deform in a controlled and predictable manner so that kinetic energy is dissipated progressively. When designed well, the crumple zone reduces the magnitude of force transmitted to the safety cell, thereby lowering the risk of injury and improving survival space. Achieving this balance depends heavily on how the structure collapses, how steadily it absorbs energy, and how effectively it avoids unstable failure modes that can lead to force spikes.

Conventional energy-absorbing components frequently utilise sandwich panels incorporating honeycomb cores. These are popular because they offer high stiffness-to-weight ratios, good structural efficiency, and relatively straightforward manufacturability. In many quasi-static scenarios, honeycomb-based panels can perform well. However, under dynamic impact loading—such as the high strain-rate conditions seen in real crash events—these structures can exhibit shortcomings that limit their effectiveness. Common issues include high initial peak impact forces, non-uniform deformation patterns, and unstable collapse behaviour such as local buckling, sudden folding, or shear band formation. In crash energy management, these behaviours are undesirable because they can lead to poor force control, less predictable deformation, and reduced energy absorption efficiency over the available crush distance. These limitations highlight the need for next-generation structural concepts that deliver stable, progressive collapse while maintaining low mass and packaging feasibility.

In response to these challenges, bio-inspired cellular structures have attracted significant attention across engineering disciplines. Nature often achieves exceptional performance through geometry—using patterns that distribute loads efficiently and manage damage gracefully. Among the most promising bio-inspired concepts are Voronoi structures, which are derived from natural tessellation patterns found in materials and biological systems. Unlike regular lattices or uniform honeycomb cells, Voronoi geometries are irregular, featuring varying cell sizes and shapes. This irregularity can encourage more uniform stress distribution, delay the onset of catastrophic local failure, and promote progressive collapse mechanisms under compressive or impact loading. As a result, Voronoi-based cellular structures are increasingly viewed as strong candidates for improving crashworthiness in lightweight applications.

That said, standalone cellular structures, including Voronoi lattices, can still face practical limitations when subjected to severe impact conditions. Localised instabilities may develop in weaker regions of the lattice, and premature densification can occur when cells collapse too quickly, shortening the effective stroke over which energy can be absorbed. If densification happens early, the structure may rapidly stiffen, driving up the transmitted loads and reducing the overall benefit to occupant protection. Therefore, while Voronoi structures offer compelling advantages, there remains a need to enhance their stability and energy absorption performance in demanding crash scenarios.

To address these concerns, this project proposes a hybrid foam-filled Voronoi sandwich panel that combines the benefits of both Voronoi cellular layers and foam core materials. In this concept, the Voronoi layers serve as the primary load-bearing and deformation-controlling components, providing the structural framework that guides collapse. The foam core complements this behaviour by contributing additional energy absorption through compression and densification, while also supporting the lattice against local buckling and reducing the likelihood of premature, localised collapse. By integrating these two mechanisms—cellular deformation in the Voronoi layers and volumetric compression within the foam—the hybrid design aims to deliver a smoother, more progressive force–displacement response.

This hybrid approach is expected to achieve several key improvements relevant to automotive crash performance. These include higher energy absorption efficiency over the crush distance, reduced peak impact forces (particularly at the initial stages of impact), and more stable, repeatable deformation behaviour. Importantly, a more controlled collapse mode can help maintain predictable crash performance across different loading conditions, improving robustness and making the concept more suitable for real-world crumple zone integration.

The study focuses on the design, optimisation, and impact analysis of the proposed hybrid structure using numerical simulation techniques. Voronoi geometries are generated and then systematically refined using the Taguchi design of experiments method, enabling efficient exploration of design variables and identification of influential parameters. Crash performance is evaluated through finite element analysis under dynamic loading conditions, allowing key indicators—such as peak force, mean crushing force, specific energy absorption, and deformation stability—to be compared across candidate designs. The intended outcome is to demonstrate the feasibility and performance potential of hybrid Voronoi–foam sandwich panels as advanced energy-absorbing structures for automotive crumple zone applications, supporting safer, lighter, and more efficient vehicle designs

The effectiveness of crumple zones, as shown in fig 1.1, lies in their ability to redistribute kinetic energy and manage deceleration. As structural components fold and compress, kinetic energy is converted into heat and sound rather than being transferred to the human body. Modern vehicles integrate crumple zones with a reinforced passenger compartment known as the safety cell, which is designed to remain intact during impact. Engineers use controlled deformation techniques, such as trigger points and strategically weakened sections, to ensure that the vehicle collapses in a predictable pattern while preventing intrusion into the cabin.

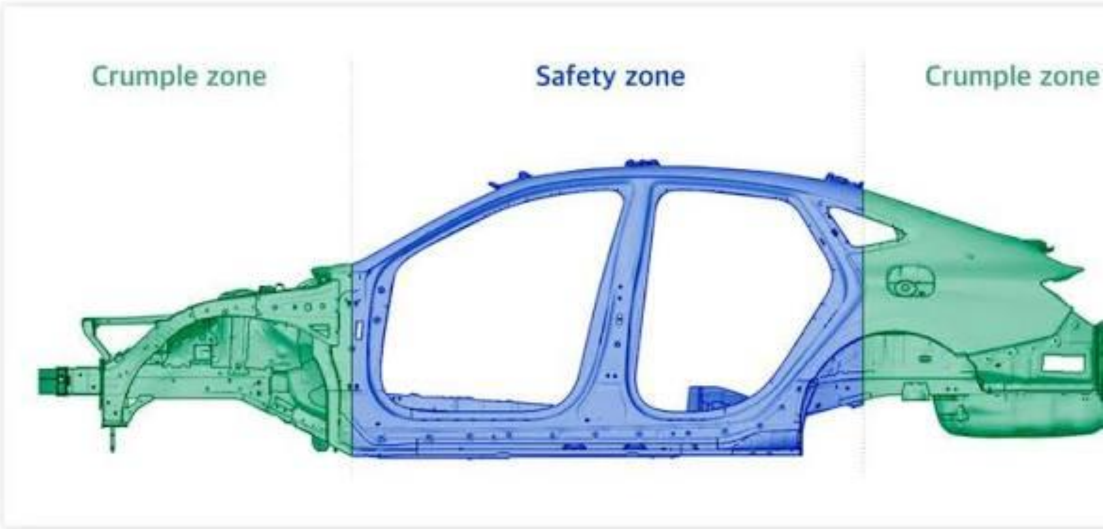

Figure 1.1: Crumple zones & Safety zone

1.1 PROBLEM STATEMENT

In modern automotive design, ensuring passenger safety during collisions while maintaining a lightweight vehicle structure remains a major challenge. Conventional energy-absorbing structures, particularly sandwich panels with uniform honeycomb cores, are widely used due to their high stiffness-to-weight ratio. However, these structures often exhibit high peak impact forces, non-uniform deformation, and unstable collapse behaviour when subjected to dynamic loading conditions.

Such characteristics lead to inefficient energy absorption and sudden transmission of forces, which can compromise the effectiveness of crumple zones and increase the risk of injury to occupants. Additionally, traditional honeycomb structures tend to show anisotropic behaviour and localized failure, limiting their performance under real-world crash scenarios.

Although bio-inspired cellular structures such as Voronoi geometries have shown potential in improving stress distribution and promoting progressive collapse, their application in hybrid sandwich structures with foam cores has not been extensively studied. Furthermore, there is a lack of systematic design optimization approaches to enhance their crashworthiness performance.

Therefore, there is a need to develop an advanced lightweight energy-absorbing structure that can:

- Reduce peak impact forces
- Improve energy absorption capacity
- Ensure stable and progressive deformation

This project addresses this gap by proposing a hybrid foam-filled Voronoi sandwich panel and evaluating its performance under impact loading conditions for application in automotive crumple zones.

1.2 OBJECTIVE

The main goal of this project is to create and test a hybrid sandwich panel made of Voronoi cellular sheets and foam layers. This design aims to improve crash energy absorption and boost crash safety in automotive crumple zone applications.

- To design Voronoi cellular sheets inspired by nature, focusing on geometric features such as cell shape, seed layout, and strut design using Rhino & Grasshopper Software.
- To use the Taguchi method for experiments, creating nine different Voronoi sheet configurations. This will help efficiently study how geometric features affect structural performance.
- To conduct static structural analysis on the designed Voronoi configurations to:
    i. Examine load distribution and deformation patterns.
    ii. Find the most structurally efficient Voronoi design.
- To choose appropriate materials for the Voronoi sheets and foam core, based on mechanical performance and how they fit with automotive safety needs.
- To develop a hybrid sandwich structure with face sheets and a core made of Voronoi and foam layers.
- To perform dynamic finite element analysis under impact conditions that mimic automotive crumple zones.
- To assess important crash safety indicators, including:
    i. Energy absorption (EA)
    ii. Specific energy absorption (SEA)
    iii. Peak crushing force
    iv. Force-displacement response
    v. Progressive collapse behavior.

1.3 SCOPE

This project focuses on evaluating the feasibility of a hybrid foam-filled Voronoi sandwich panel as an energy-absorbing structure for automotive crumple zone applications. The study is limited to the design, modelling, optimization, and numerical analysis of the proposed structure using computational tools. The scope includes the development of Voronoi cellular geometries using Rhino software, followed by the application of the Taguchi design of experiments method to generate multiple design configurations. The analysis is carried out to identify an optimal combination of geometric parameters such as cell count, seed distribution, and strut thickness factor. The project involves the creation of a multi-layer sandwich panel consisting of Voronoi sheets and an aluminium foam core. The structural performance of the developed model is evaluated through finite element simulations using ANSYS and ABAQUS under controlled loading conditions. The study considers different loading scenarios, including drop ball impact analysis and plate crush analysis, to simulate real-world crash conditions. The scope further includes the evaluation of key crashworthiness parameters such as:

- Energy Absorption (EA)
- Specific Energy Absorption (SEA)
- Peak Crushing Force (PCF)
- Mean Crushing Force (MCF)
- Crushing Force Efficiency (CFE)

The study is limited to numerical simulation-based analysis and does not include physical experimentation or prototype fabrication. Material behavior is considered based on predefined properties, and the internal microstructure of the foam is not explicitly modeled to maintain computational efficiency.

Economic feasibility, manufacturing considerations, and long-term durability of the proposed structure are beyond the scope of this project. The work primarily aims to assess the structural performance and crashworthiness potential of the hybrid Voronoi–foam sandwich panel.

1.4 MOTIVATION

The main reason for this project is to create lightweight, efficient, and reliable energy-absorbing structures for modern vehicles without depending only on traditional uniform-core sandwich designs. Even though conventional honeycomb and foam-core sandwich panels are common, they often have sudden force spikes and premature densification. This can transfer high loads to the vehicle cabin and endanger occupant safety.

Recent progress in computational modelling and additive manufacturing has opened up possibilities for designing complex cellular shapes, like Voronoi structures, that were hard to develop and study before. These structures have natural geometric irregularities, which can improve energy absorption by preventing synchronized collapse and promoting gradual deformation. However, standalone Voronoi cellular structures might face local instability or excessive buckling under heavy axial loads.

Combining foam infill with Voronoi cellular sheets offers a good solution. It stabilizes cellular struts, lowers peak crushing forces, and adds more energy absorption through foam crushing and densification. Additionally, using design of experiments (DOE) techniques, like the Taguchi method, allows for systematic exploration of various design parameters with less computational effort. This makes the optimization process more efficient and organized. This project is therefore driven by the need to:

- Improve crash energy absorption efficiency
- Achieve controlled progressive collapse under axial impact
- Reduce structural weight
- Explore bio-inspired design concepts for automotive safety applications

CHAPTER 2

LITERATURE SURVEY

A substantial body of research has examined the crashworthiness and impact performance of sandwich structures, with particular attention given to honeycomb cores and hybrid, foam-filled solutions. This interest is driven by the need to improve energy absorption, delay catastrophic failure, and achieve more predictable deformation under a range of impact scenarios, from low-velocity strikes to more severe ballistic or crash-type loading. Across the literature, a consistent theme emerges: the core architecture and any internal filling material strongly influence how effectively a sandwich panel manages impact energy and maintains structural integrity.

Hassanpour Roudbeneh et al. (2019) carried out an experimental study on honeycomb sandwich panels filled with polyurethane foam and subjected to impact loading. Their work demonstrated that foam filling can markedly increase dynamic strength and boost energy absorption when compared with unfilled honeycomb structures. In practical terms, the foam acts to support the honeycomb cell walls, restrain local buckling and distribute stresses more evenly across the core. The authors also reported that raising the foam density further improves performance, delivering higher ballistic limits and greater specific energy absorption. In some cases, these gains reached approximately 24% relative to unfilled panels, underlining the potential of foam density as a straightforward design lever for enhanced protection.

Topkaya and Solmaz (2018) focused on low-velocity impact behaviour in honeycomb sandwich composites, highlighting the importance of key structural parameters. Their findings indicated that face sheet thickness is a dominant factor in governing impact resistance, since the facesheets are responsible for initial contact, load distribution and preventing penetration. By contrast, core thickness showed a comparatively limited effect within the conditions studied, suggesting that simply increasing core depth may not be as effective as optimising the facesheets when the goal is to resist surface damage and local indentation. Overall, their study reinforces the broader design message that targeted parameter selection—rather than uniform material increases—can deliver more efficient improvements in impact performance.

Fan et al. (2023) investigated Nomex honeycomb sandwich panels under varying impactor conditions, providing insight into how loading configuration changes the observed damage and energy absorption response. Their results showed that both core type and impactor shape significantly influence deformation mechanisms, including the extent of local crushing, facesheet damage and the evolution of core collapse. This is particularly relevant for real-world applications, where impact threats are diverse and may range from blunt objects to sharper geometries that concentrate stresses. The study therefore points to the need for performance validation across multiple impact scenarios, not only a single test condition.

Valente et al. (2023) explored aluminium foam-filled honeycomb crash absorbers, with an emphasis on design choices and failure mechanisms. Their work highlighted that combining foam with honeycomb architecture can improve crashworthiness by promoting more stable, progressive deformation. Rather than exhibiting abrupt collapse or highly localised failure, the hybrid system can dissipate energy more consistently as both the foam and the honeycomb contribute to resistance. This stability is crucial in crash absorbers, where controlled crushing is often preferred to maximise energy dissipation while limiting peak loads transmitted to surrounding structures or occupants.

Tao et al. (2022) conducted a comparative study of foam-filled and honeycomb-filled thin-walled structures, demonstrating that hybrid configurations can outperform single-material fillings. The reported improvements in energy absorption and crash performance were attributed to the synergistic interaction between foam and cellular structures: the foam supports and stabilises the cellular walls, while the cellular framework helps constrain the foam, encouraging effective densification and sustained load capacity throughout deformation. These complementary roles help explain why multi-phase designs are increasingly attractive for protective structures.

Beyond these individual studies, more recent research on composite sandwich structures continues to show that incorporating foam within cellular cores can raise load-carrying capacity and improve impact resistance. The interaction between foam and cell geometry tends to enhance progressive collapse behaviour, limiting the likelihood of sudden, brittle failure modes and improving the predictability of damage evolution. Such behaviour is valuable not only for safety, but also for maintenance planning, since gradual failure mechanisms are often easier to detect and manage than abrupt fractures.

Finally, optimisation studies consistently emphasise that geometric parameters strongly govern energy absorption and deformation under impact loading. Variables such as cell shape, wall thickness, relative density, core configuration, and the arrangement of hybrid regions can all shift the balance between stiffness, crush strength and energy dissipation. As a result, effective crashworthy design typically involves an integrated approach: selecting appropriate materials, tailoring facesheet and core dimensions, and refining cellular geometry to match the expected impact conditions and performance targets.

CHAPTER 3

DESIGN OF VORONOI STRUCTURE

3.1 INTRODUCTION TO VORONOI

In mathematics, a Voronoi diagram, as shown in fig 3.1, is a partition of a plane into regions close to each of a given set of objects. It can be classified also as a tessellation. In the simplest case, these objects are just finitely many points in the plane (called seeds, sites, or generators). For each seed there is a corresponding region, called a Voronoi cell, consisting of all points of the plane closer to that seed than to any other. The Voronoi diagram of a set of points is dual to that set's Delaunay triangulation.

The Voronoi diagram is named after mathematician Georgy Voronoy, and is also called a Voronoi tessellation, a Voronoi decomposition, a Voronoi partition, or a Dirichlet tessellation (after Peter Gustav Lejeune Dirichlet). Voronoi diagrams have practical and theoretical applications in many fields, mainly in science and technology, but also in visual art. Unlike traditional honeycomb structures, Voronoi cells provide:

- Improved stress distribution
- Progressive deformation
- Reduced stress concentration

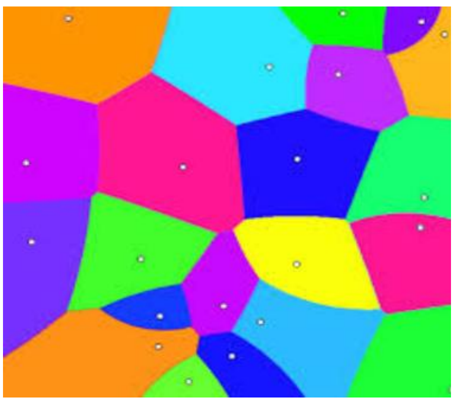

Figure 3.1 Voronoi Structure

3.2 DESIGN OF VORONOI SHEET

The Voronoi cellular structures were generated by placing stochastic (randomised) seed points throughout the design domain, producing an irregular network of polyhedral cell geometries. This workflow was developed in Rhino, using the Grasshopper add-on, which is shown in fig 3.2, to control and iterate the distribution, spacing, and overall morphology of the cells. By adopting a stochastic topology rather than a uniform lattice, the structure promotes more effective load redistribution as deformation progresses. In practical terms, this helps minimise the likelihood of localised stress concentrations that can trigger premature failure. As a result, the cellular architecture is better able to support a stable, progressive collapse mechanism under impact loading, improving energy absorption and overall robustness.

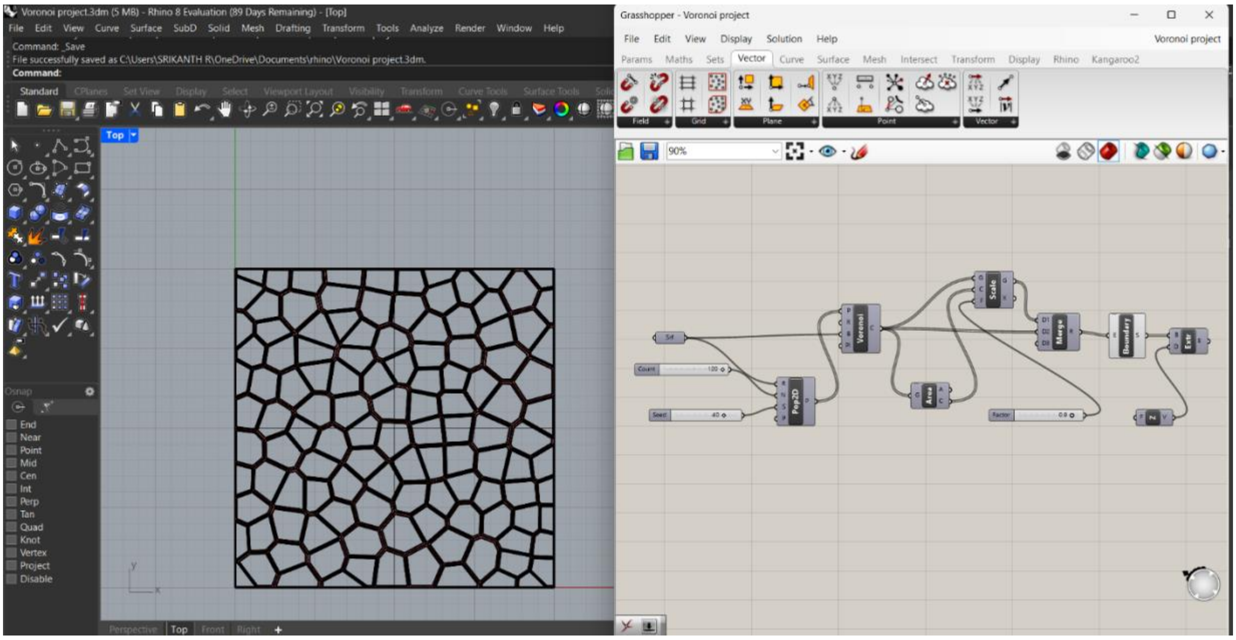

Figure 3.2: Voronoi Sheet formation using Rhino & Grasshopper

CHAPTER 4

OPTIMIZATION OF VORONOI SHEET

4.1 TAGUCHI DESIGN OF EXPERIMENTS

To methodically examine how Voronoi geometric variables affect the crashworthiness behavior of the sandwich structure, the Taguchi design of experiments (DOE) technique was used. The Taguchi method offers a practical statistical strategy for studying the impact of several design factors while greatly lowering the number of simulations needed compared with a traditional full factorial plan. As a result, parameter interactions can be investigated and the best design settings can be identified with limited computational cost.

In this work, three primary geometry-related factors defining the Voronoi cellular architecture were chosen for optimization: cell count, seed distribution, and strut thickness factor. Each factor was evaluated at three separate levels to represent changes in the Voronoi lattice geometry and to observe how these changes influence the structural response. The cell count determines the tessellation density of the Voronoi pattern, and therefore affects how many load-carrying members exist in the structure. The seed distribution factor controls the degree of spatial randomness and the resulting cell topology, which in turn alters deformation modes and the way stresses are distributed. The strut thickness factor specifies the relative thickness of the lattice struts and has a direct effect on stiffness and overall load-supporting capacity within the cellular framework.

Using these three factors and three levels, as shown in table 4.1, a Taguchi L9 orthogonal array was selected to plan the numerical trials. This array cuts the simulation requirement from 27 cases in a full factorial design down to nine simulations, while preserving orthogonality and balanced representation of the factors. For every run, finite element analyses, as shown in fig 4.1 were performed to extract the maximum reaction force and the maximum directional deformation of the Voronoi sheet under the applied loading scenario.

Table 4.1 : Taguchi DOE parameters

Cell count	Seed	Strut Thickness Factor
120	10	0.8
80	30	0.8
100	50	0.8
120	30	0.85
100	10	0.85
80	50	0.85
120	50	0.9
100	30	0.9
80	10	0.9

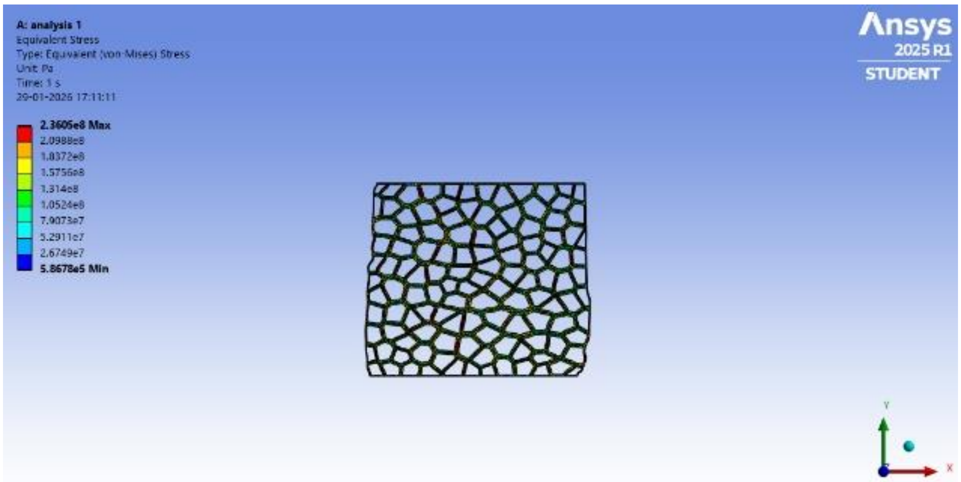

Figure 4.1: Static Structural Analysis for Voronoi Sheet

To compare the robustness and performance of each parameter combination, the signal-to-noise (S/N) ratio was computed using the larger-the-better formulation, shown in eqn 4.1, because greater peak force corresponds to stronger crashworthiness and improved load-carrying performance. The S/N metric captures both average performance and scatter, enabling the selection of parameter levels that raise structural strength while reducing sensitivity to variations.

Eqn 4.1: Larger-the-better: S/N=20log10(Max Force Reaction)

CHAPTER 5

MATERIAL SELECTION

5.1 MATERIAL FOR VORONOI SHEETS

The selected material for the Voronoi cellular sheet is Aluminium alloy Al 5052-H32, a well-regarded grade that offers an effective balance of strength, formability, and ductility. It also delivers strong resistance to corrosion, particularly in environments where moisture or road salts may be present. These combined properties make it a reliable option for lightweight structural components, helping meet performance and durability requirements commonly associated with automotive engineering and broader vehicle body or chassis applications, which is mentioned in Table 5.1.

Table 5.1: Material Property of Al-5052 used

Property	Value
Density (ρ)	2680 kg/m^3
Young's Modulus (E)	7×10^{10} Pa
Poisson's Ratio (ν)	0.33
Bulk Modulus (K)	6.86×10^{10} Pa
Shear Modulus (G)	2.63×10^{10} Pa
Yield Strength (σ_γ)	1.93×10^{8} Pa
Tangent Modulus(E_t)	1.5×10^{8} Pa

5.2 MATERIAL FOR FOAM CORE

The material selected for the core has been identified as aluminium foam, primarily due to its favourable mechanical properties and the distinctive manner in which it deforms when subjected to compressive loading. In general, foam materials are defined by their notably low density and comparatively low elastic stiffness, while offering an exceptionally high capacity to absorb and dissipate energy. These combined characteristics make them particularly well suited to applications where weight reduction and impact or crush performance is important, and they clearly set foams apart from conventional solid metals. The material property for the aluminium foam has been listed below in the table 5.2.

Table 5.2: Material Property for Aluminium foam used

Property	Value
Density (ρ)	400 kg/m³
Young's Modulus (E)	5×10^8 Pa
Poisson's Ratio (ν)	0.30
Bulk Modulus (K)	4.17×10^8 Pa
Shear Modulus (G)	1.92×10^8 Pa
Plastic Hardening (Multilinear)	
Stress (Pa)	Plastic Strain
3.0×10^6	0
3.2×10^6	0.05
3.5×10^6	0.10
4.0×10^6	0.20
6.0×10^6	0.40
8.0×10^6	0.50

These material properties were assigned to the Voronoi sheet and foam core during the computational analysis in both Ansys and Abaqus. These data values are key in computing the final values for the project.

CHAPTER 6

DESIGN OF FOAM FILLED SANDWICH PANEL

6.1 DESIGN OF FOAM

The foam core was deliberately engineered to emulate the behaviour of a closed-cell cellular material, which is widely used in structural sandwich panels because it offers an excellent combination of compressive strength, shear stiffness, and overall structural stability under demanding loading conditions. In practical applications, these characteristics help sandwich constructions maintain shape, resist buckling, and deliver reliable load transfer between the face sheets and the core.

Closed-cell foams are distinguished by their discrete, sealed pores, each one enclosed by thin solid cell walls. This isolated pore structure allows the material to withstand significant compressive loads while undergoing a progressive, controlled densification as deformation increases. Rather than collapsing all at once, the foam typically exhibits a gradual crushing response, which is particularly valuable when energy absorption and impact mitigation are important design considerations. To replicate the energy-absorbing behaviour that is typical of cellular structures, the foam core selected for the sandwich panel was chosen specifically on the basis of relevant material properties.

Instead of explicitly modelling the foam's complex internal micro-architecture (with its many individual cells), the material was represented through its bulk mechanical characteristics, including density, elastic modulus, and plastic hardening behaviour. This continuum-style description captures the essential macroscopic response without requiring a highly detailed geometric representation. Adopting this approach preserves computational efficiency during explicit dynamic analysis, where solution time and stability constraints can become critical. At the same time, it enables the simulation to reproduce key aspects of foam performance, particularly its compressive response and capacity to absorb energy through progressive deformation and densification, making it well suited for evaluating the structural and protective behaviour of the sandwich panel system.

6.2 FINAL SANDWICH PANEL

The hybrid energy-absorbing structure investigated in this study takes the form of a sandwich panel comprising two Voronoi-based cellular sheets separated by a compliant foam core, which is shown in fig 6.1. In terms of overall geometry, the core section of the sandwich has a total thickness of 22 mm. This thickness is built up from three distinct layers: two outer Voronoi layers, each measuring 6 mm thick, and a centrally positioned foam layer with a thickness of 10 mm. The panel is manufactured with a square planform, providing a consistent footprint of 200 mm × 200 mm, which also supports repeatable loading conditions during testing or simulation.

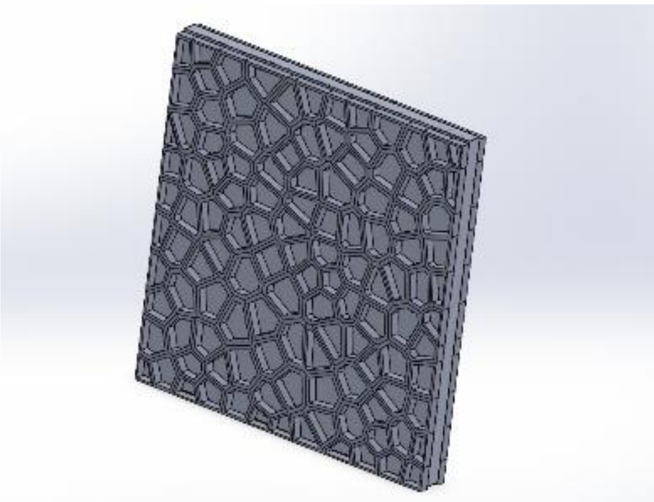

Figure 6.1: Voronoi Sandwich Panel

Functionally, the Voronoi cellular sheets act as the primary load-carrying components within the sandwich architecture. Under compressive loading, they are designed to initiate and sustain progressive deformation rather than failing abruptly. This controlled collapse mechanism is critical for impact mitigation, as it promotes stable energy dissipation over a larger displacement. A significant proportion of the absorbed energy is attributed to bending-dominated behaviour and plastic deformation within the lattice struts, where the irregular Voronoi topology helps distribute stresses and encourages multiple deformation zones to develop across the sheet.

Positioned between these two cellular skins, the foam core contributes additional performance benefits that improve the overall crashworthiness of the panel. Beyond simply separating the Voronoi layers, the foam provides a secondary energy-absorption pathway through progressive compression of the cellular material. As loading increases, the foam undergoes compaction and eventual densification, converting impact work into deformation while helping to stabilise the crushing response. Working together, the Voronoi sheets and foam core create a hybrid system that balances load support, controlled collapse, and efficient energy absorption for compressive impact scenarios.

CHAPTER 7

NUMERICAL ANALYSIS MODEL

Finite element analysis was performed to assess the structural behavior of the hybrid sandwich panel when subjected to compressive loading conditions. Three-dimensional models of the Voronoi cellular sheets and foam core were created using Solidworks software and subsequently imported into the Ansys Explicit and Abaqus for simulation.

An impact speed of 20 m/s (about 72 km/hr) was selected to replicate a realistic vehicle crash condition commonly applied in crashworthiness studies. This speed enables evaluation of the structure's energy absorption capacity and deformation response when subjected to intense dynamic impact loading.

7.1 DROP BALL ANALYSIS

Drop ball analysis is a widely used approach for simulating high-velocity impact events and understanding how a structure behaves under sudden, short-duration loading. By modelling a controlled impact scenario, it enables engineers to evaluate key responses such as overall deformation, localised stress distribution, and the structure's ability to absorb and dissipate kinetic energy through progressive damage or plastic deformation. In practical terms, the analysis helps determine whether the component can withstand impact without catastrophic failure, and whether it can manage the incoming energy in a stable and predictable way under explicit dynamic conditions.

In this study, a drop ball impact simulation was performed to investigate the surface impact performance of the sandwich panel, which is shown in fig 7.2. Here, the Voronoi sandwich structure was subjected to impact loading on its face sheet, driven by a spherical impactor with an initial velocity of 20 m/s. The impact direction was applied parallel to the sandwich panel's face, allowing the simulation to capture how the surface layers respond to high-speed contact, including indentation behaviour, stress waves travelling through the face sheet, and the transfer of load into the cellular core.

In addition, a separate explicit dynamic drop ball analysis was carried out to assess the axial impact performance of the Voronoi sandwich panel, which is shown in fig 7.1. In this case, the core region of the Voronoi structure was directly impacted by a spherical impactor, again with the same initial velocity of 20 m/s. The impactor was oriented perpendicular to the sandwich panel's axis, so the model could evaluate how the core crushes, how stresses develop through the thickness, and how efficiently the structure absorbs energy under axial loading.

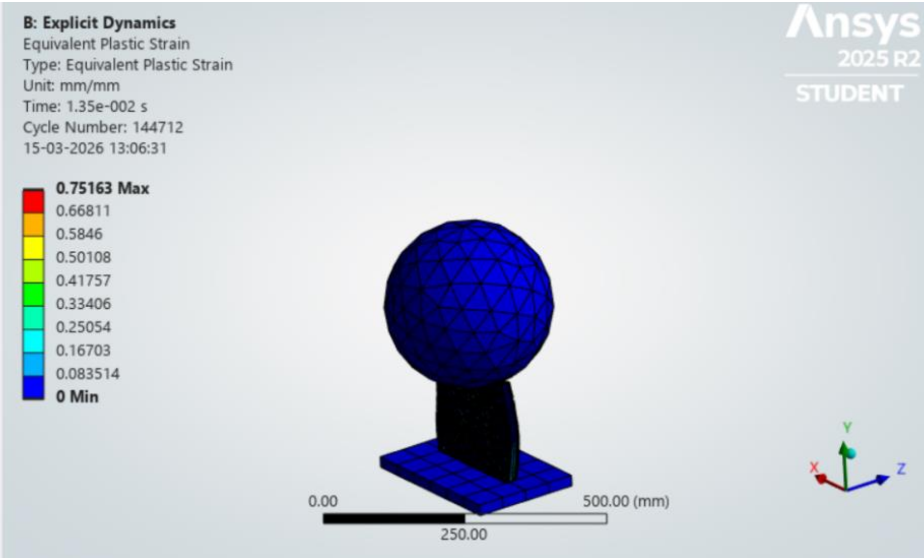

Figure 7.1: Axial Drop Ball Analysis

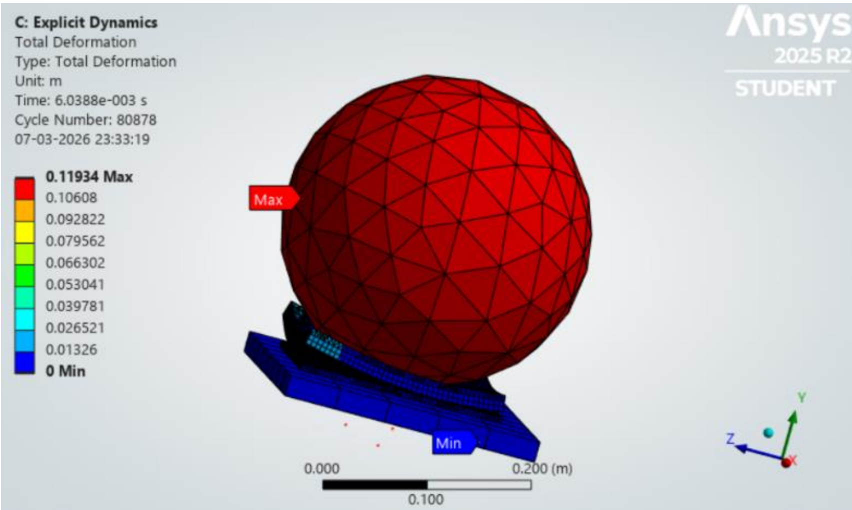

Figure 7.2: Surface Drop Ball Analysis

For consistency and to replicate a rigid backing condition, the plate beneath the sandwich panel was defined with fixed support boundary conditions in both simulations. This constraint ensures that differences in the results primarily reflect changes in impact orientation and load path, rather than variations in overall support or movement of the base. The analysis details are listed in table 7.1.

Table 7.1: Analysis Details for Ansys Explicit Drop Ball Impact Analysis

Parameter	**Description**
Mass of the Impactor	110.87 kg
Velocity	20 m/s
Element Shape	Tetrahedral
Element Size in Sandwich Structure	5 mm
Total Number of Elements	23,558
Global Element size	21.556 mm
End Time	0.015 s

7.2 PLATE CRUSH ANALYSIS

The plate crush analysis, as shown in fig 7.3, was carried out to characterise the compressive response and energy absorption capability of the Voronoi sandwich panel under controlled crushing loads. This type of simulation is particularly valuable because it reveals how the panel behaves under progressively increasing compressive forces, providing essential insight into crashworthiness, structural integrity, and failure progression. By monitoring key outputs such as load–displacement behaviour, deformation patterns, and absorbed energy, the analysis helps determine whether the sandwich concept can deliver stable crushing performance and effective impact mitigation in demanding applications.

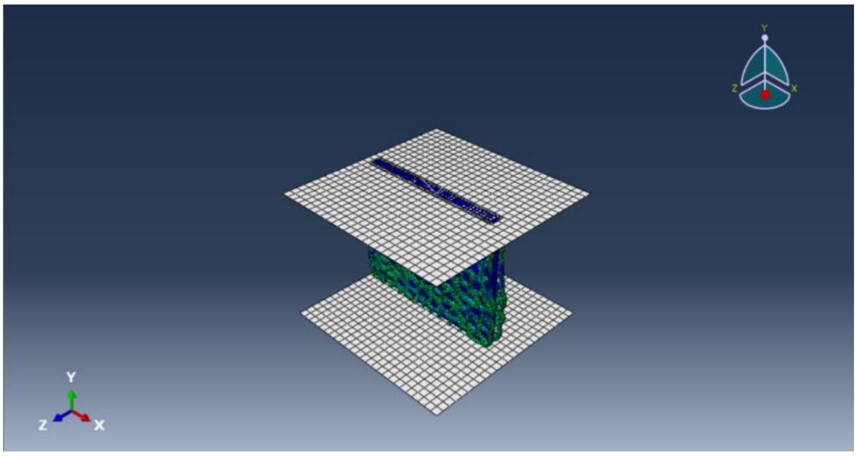

Figure 7.3: Plate Crush Analysis in Abaqus

Table 7.2: Analysis Details for Abaqus Plate Crush Analysis

Parameter	**Description**
Element Type	C3D8R
Element Shape	Hexahedral elements
Velocity	20 m/s
Global Element Size	3 mm
Total Number of Elements	~30,000
Mesh Convergence Check	Performed to ensure mesh-independent results
End Time	0.015 s

For the axial crushing configuration, a rigid plate was positioned perpendicular to the sandwich panel and driven along the panel's normal direction. This created through-thickness compression across the full depth of the structure, replicating conditions where a panel is squeezed between two surfaces. During the crush event, the applied stress was first carried by the upper face sheet, which initially resists and distributes the load. As crushing continued, the load was transmitted into the foam layer and then into the Voronoi lattice core, where progressive deformation and cell collapse mechanisms contribute significantly to energy dissipation. This sequential transfer of load through each layer is critical for understanding which component dominates stiffness, which initiates failure, and how the core supports stable crushing. The analysis details for the plate crush in Abaqus is listed in table 7.2.

CHAPTER 8

RESULT AND DISCUSSION

8.1 VORONOI SHEET OPTIMIZATION

ANOVA, table 8.1, shows strut thickness is the most influential factor on maximum force, with the highest F-value and a very low p-value confirming strong statistical significance. Cell count also meaningfully affects load-bearing capacity by changing Voronoi lattice density, while seed distribution has a smaller impact despite influencing geometric randomness.

Table 8.1: ANOVA for the varying parameters of Voronoi

Source	**Degrees of Freedom (DF)**	**Sum of Squares (SS)**	**Mean Square (MS)**	**F-Value**	**P-Value**
Cell Count	2	0.872	0.436	27.05	0.0356
Seed	2	0.211	0.105	6.54	0.1326
Strut Thickness	2	137.052	68.526	4251.21	0.0002
Error	2	0.032	0.016	—	—
Total	**8**	**138.167**	—	—	—

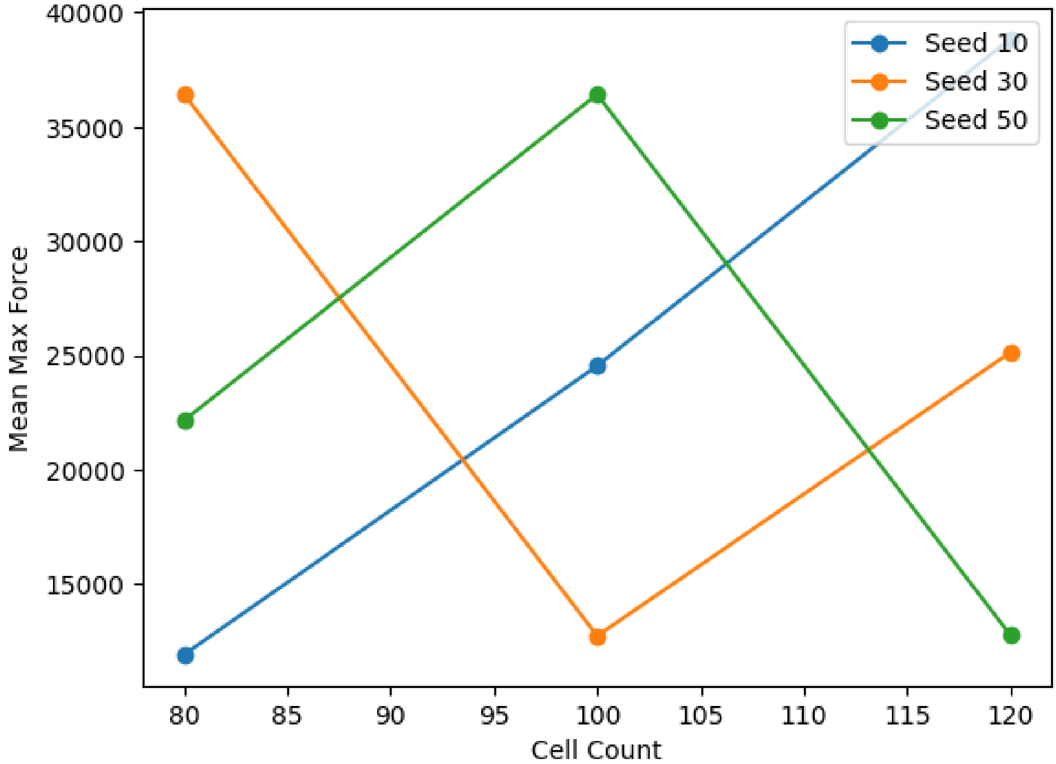

Figure 8.1: Interaction Plot between Cell Count and Seed

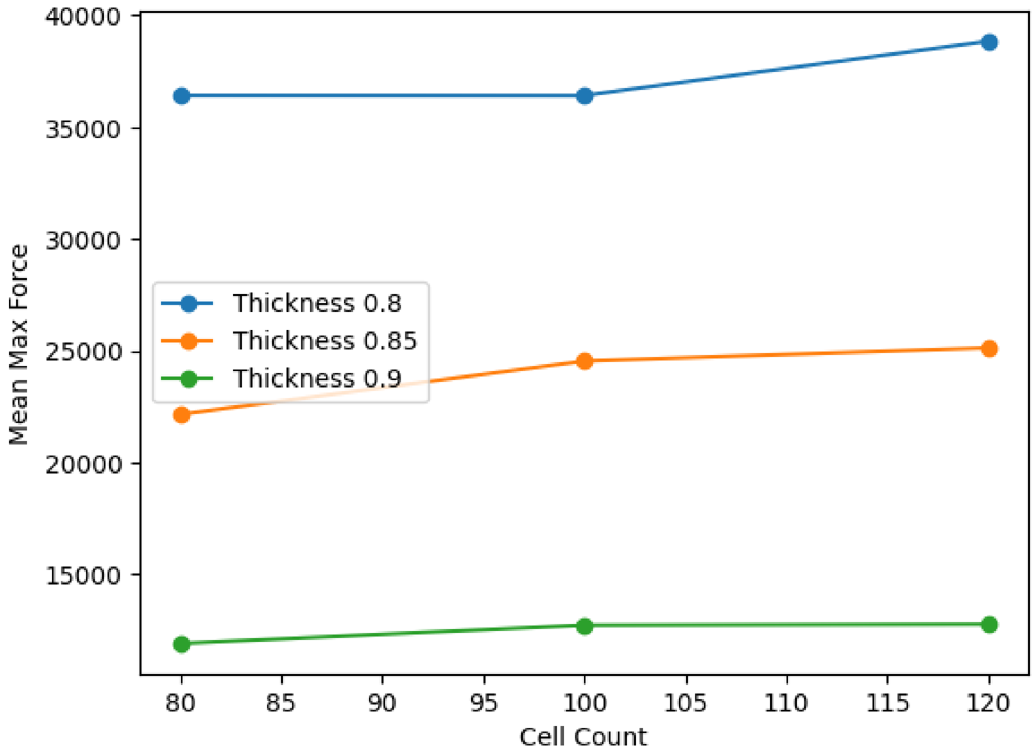

Figure 8.2: Interaction Plot between Cell Count and Strut thickness factor

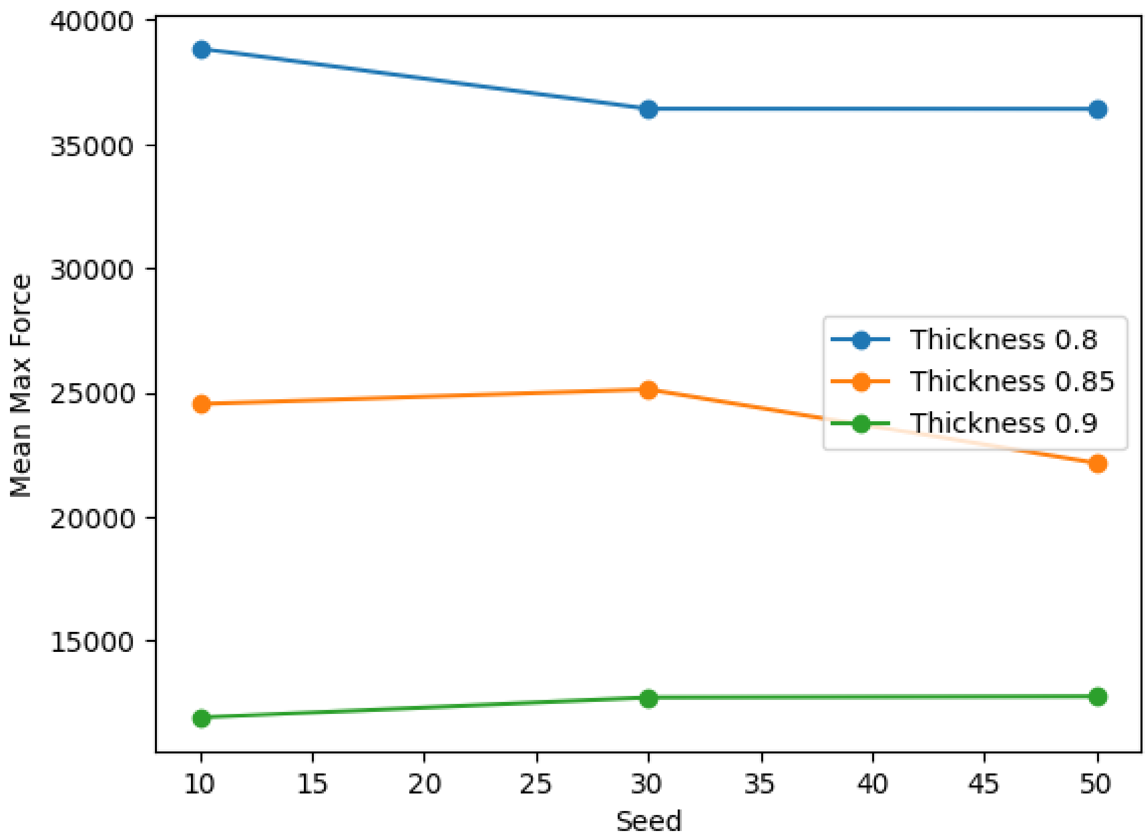

Figure 8.3: Interaction Plot between Seed and Strut thickness factor

From the cell count–seed interaction plot, fig 8.1, the non-parallel and intersecting lines indicate a significant interaction between these two parameters, suggesting that the influence of seed distribution on the maximum force varies with cell count. In contrast, the cell count–strut thickness factor plot, fig 8.2, exhibits nearly parallel lines, indicating minimal interaction and confirming that strut thickness independently influences the structural response. Similarly, the seed–strut thickness interaction plot, fig 8.3, shows only slight deviations from parallelism, implying a weak interaction between these parameters.

Overall, the results suggest that while cell count and seed distribution exhibit some level of interaction, the strut thickness factor remains the dominant parameter, with its effect being largely independent of the other variables. This observation is consistent with the ANOVA results, which identified strut thickness as the most significant factor affecting the maximum force.

Table 8.2: Taguchi Implementation and the results obtained in Ansys

Cell count	Seed	Strut Thickness Factor	Max Force Reaction (N)	Max Directional Deformation (m)	SN Ratio (dB)
120	10	0.8	38837	0.00037686	91.7849135
80	30	0.8	36419	0.031788	91.22656033
100	50	0.8	36412	0.030279	91.22489068
120	30	0.85	25123	0.0015714	88.00142997
100	10	0.85	24537	0.00068038	87.79642926
80	50	0.85	22153	0.00023111	86.90865095
120	50	0.9	12767	0.00078656	82.12177717
100	30	0.9	12725	0.001107	82.09315582
80	10	0.9	11910	0.03051	81.51823523

The Taguchi method was employed to determine the optimal combination of Voronoi geometric parameters for maximizing the crashworthiness performance of the sandwich structure. Based on the signal-to-noise (S/N) ratio analysis using the larger-the-better criterion, the optimal levels of the design parameters were identified.

The response table for S/N ratios indicated that the optimal parameter combination corresponds to:

- Cell Count = 120 (Level 3)
- Seed Distribution = 30 (Level 2)
- Strut Thickness Factor = 0.8 (Level 1)

This combination yielded the highest S/N ratio, indicating superior performance in terms of maximum force response and structural robustness, which is shown in table 8.2.

8.2 PARAMETERS OBTAINED

The key crashworthiness parameters considered in this project are widely used to evaluate the energy absorption capability and structural performance of sandwich panels under impact and crushing conditions. The associated parameters are given below:

(a) Energy Absorption (EA):

Energy Absorption (EA) refers to the total mechanical energy a structure can take in while it deforms under impact or crushing-type loads. In practical terms, it reflects the structure's capacity to dissipate incoming kinetic energy, primarily through plastic deformation rather than elastic rebound. A higher EA generally indicates better crashworthiness because more energy is consumed in controlled deformation. Mathematically, energy absorption is obtained by integrating the crushing force over the deformation distance (eqn.8.1).

$$\textbf{Eqn 8.1 : } EA = \int F \cdot dx$$

(b) Specific Energy Absorption (SEA):

Specific Energy Absorption (eqn 8.2) describes how much energy is absorbed per unit mass of the structure. Because weight is critical in many applications (such as automotive, aerospace, and protective systems), SEA is widely used to compare the energy-absorbing efficiency of different materials and designs on an equal-mass basis. A structure with high SEA can provide strong impact protection while remaining lightweight.

$$\textbf{Eqn 8.2 : } SEA = EA / m$$

(c) Peak Crushing Force (PCF):

Peak Crushing Force is the highest force recorded during crushing or impact. This value is important because excessive peaks can lead to sudden load transfer, increasing the risk of damage or injury in real-world crash scenarios. PCF is typically taken as the maximum force on the force–displacement curve (eqn 8.3).

$$\textbf{Eqn 8.3 : } PCF = F_{max}$$

(d) Mean Crushing Force (MCF):

Mean Crushing Force represents the average force sustained during progressive crushing over a given crush distance. It provides a useful measure of the overall load level maintained while absorbing energy and is often used to assess the stability of the crushing process (eqn 8.4).

$$\textbf{Eqn 8.4 : MCF = EA / } \boldsymbol{\delta}$$

(e) Crushing Force Efficiency (CFE):

Crushing Force Efficiency indicates how uniformly the structure crushes by comparing the average crushing force to the peak value. A higher CFE suggests a smoother, more stable response with fewer extreme force spikes, which is typically desirable for safer and more predictable energy absorption(eqn 8.5).

$$\textbf{Eqn 8.5 : CFE = MCF / PCF}$$

8.3 DROP BALL TEST RESULTS

The performance of the hybrid foam-filled Voronoi sandwich panel under drop ball impact loading was evaluated for both surface impact and axial impact conditions. The results are compared based on key crashworthiness parameters such as Energy Absorption (EA), Specific Energy Absorption (SEA), Mean Crushing Force (MCF), Peak Crushing Force (PCF), and Crushing Force Efficiency (CFE), which is shown in table 8.3.

Energy Absorption (EA)

The axial impact condition exhibited significantly higher energy absorption (6521.6 J) compared to surface impact (3812.2 J). This indicates that the structure is more efficient in dissipating energy when the load is applied along the axial direction, due to better engagement of the cellular structure.

Specific Energy Absorption (SEA)

The SEA value for axial impact (10464.7 J/kg) is considerably higher than that of surface impact (6116.6 J/kg). This suggests that the structure provides superior energy absorption per unit mass under axial loading, making it more efficient for lightweight crashworthy applications.

Mean Crushing Force (MCF)

The mean crushing force is higher in axial impact (96,900 N) compared to surface impact (38,900 N). This indicates a more stable and sustained load-carrying capacity during deformation in the axial direction.

Peak Crushing Force (PCF)

The peak crushing force for surface impact (1.008×10^6 N) is much higher than that of axial impact (1.647×10^5 N). A higher peak force is undesirable in crashworthiness applications, as it can transmit excessive force to passengers or protected components. Therefore, axial impact shows a safer response due to its lower peak force.

Crushing Force Efficiency (CFE)

The CFE value is significantly higher for axial impact (0.583) compared to surface impact (0.0386). This indicates that axial loading results in more uniform force distribution and progressive deformation, which is ideal for energy-absorbing structures.

Overall Discussion

From the drop ball test results, it is evident that the hybrid Voronoi sandwich structure performs significantly better under axial impact conditions. The axial loading promotes progressive collapse, higher energy absorption, and lower peak forces, which are desirable characteristics for crashworthy structures. In contrast, surface impact leads to high peak forces and poor energy absorption efficiency, indicating localized failure and unstable deformation.

Thus, the structure is more suitable for applications where axial loading dominates, such as in automotive crumple zones and impact protection systems.

Table 8.3: Results of Drop Ball Impact Analysis

Parameter	Drop Ball	
	Surface Impact	Axial Impact
Energy Absorption (J)	3812.2	6521.6
Specific Energy Absorption (J/kg)	6116.6	10464.7
Mean Crushing Force (N)	38900	96900
Peak Crushing Force (N)	1.008×10^{6}	1.647×10^{5}
Crushing Force Efficiency (CFE)	0.0386	0.583

8.4 PLATE CRUSH TEST RESULTS

The crushing behaviour of the hybrid foam-filled Voronoi sandwich panel under axial plate loading was investigated to better understand its deformation modes and overall energy-absorbing capability. The aim of this evaluation was to characterise how the structure responds when compressed in a quasi-static, plate-driven manner, and to quantify its crashworthiness using widely adopted metrics: Energy Absorption (EA), Specific Energy Absorption (SEA), Mean Crushing Force (MCF), Peak Crushing Force (PCF), and Crushing Force Efficiency (CFE), which is listed on the table 8.4. Together, these parameters provide a clear picture of not only how much energy the panel can dissipate, but also how stable, predictable, and design-friendly the crushing response is under sustained compressive loading.

Energy Absorption (EA)

During axial plate crushing, the hybrid panel demonstrates effective energy absorption, largely because the applied load promotes relatively uniform compression of both the Voronoi cellular architecture and the supporting foam core.

With a plate load, the force is introduced across a broad area, encouraging load sharing throughout the cross-section rather than concentrating stresses at a single point. As crushing progresses, the Voronoi cell walls progressively buckle, fold, and compact, while the foam core contributes by resisting collapse, reducing local instability, and providing continuous support to the surrounding lattice. This combined mechanism leads to steady, progressive deformation and sustained energy dissipation over the crushing stroke.

In contrast to impact-dominated events, where inertia and strain-rate effects can trigger abrupt local failures, plate crushing typically produces a more controlled and gradual response. The energy is absorbed through a sequence of stable collapse events rather than a single dominant failure mode, which is beneficial for applications where predictable behaviour and controlled load transfer are priorities. Overall, the EA response under plate loading indicates that the structure can dissipate significant energy without relying on highly localised damage, helping maintain structural integrity as deformation accumulates.

Specific Energy Absorption (SEA)

Specific Energy Absorption (SEA), which normalises energy absorption by mass, indicates a moderate level of efficiency under axial plate crushing. This outcome is consistent with quasi-static crushing conditions, where the structure is not able to “benefit” from additional dynamic strengthening mechanisms that can occur in higher velocity scenarios. Even so, the SEA performance remains meaningful because it reflects the panel’s capacity to absorb energy consistently on a per-mass basis, which is essential for weight-sensitive design targets.

Although SEA may be slightly lower than results observed under high-velocity impact cases, the plate crushing response highlights a different advantage: repeatability and controlled behaviour. In practical terms, the panel can still offer attractive energy dissipation while maintaining a stable crushing pattern, and this can be more valuable in certain engineering contexts than maximising SEA alone. The presence of the foam core is particularly relevant here, as it helps prevent premature localisation of deformation, encouraging broader participation of the cellular network and supporting more consistent energy absorption.

Mean Crushing Force (MCF)

The mean crushing force (MCF) remains relatively steady throughout the crushing process, suggesting the structure offers uniform resistance to compression once collapse initiates. This stability is an important indicator of progressive collapse behaviour: rather than exhibiting erratic fluctuations or repeated force spikes, the panel sustains a more consistent load level as it compacts. For crashworthiness and protective design, a stable MCF is often desirable because it corresponds to predictable energy dissipation and reduces the likelihood of sudden changes in loading that could compromise adjacent components.

From a structural mechanics perspective, this behaviour implies that the Voronoi network and foam core are engaging together in a balanced way. As individual cell regions begin to fold or collapse, neighbouring regions continue to carry load, and the foam helps distribute stresses, preventing the crushing response from becoming dominated by a single weak zone. The result is a smoother force–displacement profile and a collapse mechanism that is well-suited to controlled deformation applications.

Peak Crushing Force (PCF)

The peak crushing force (PCF) observed under axial plate loading is lower than what is typically recorded under dynamic impact conditions. This reduction is generally advantageous, as high peak loads can transmit sudden forces into the surrounding structure, potentially causing damage, undesirable accelerations, or early failure in connected parts. A lower PCF suggests the panel begins to crush in a more progressive manner, with reduced initial resistance spikes, which is often a design goal for energy absorbers intended to protect occupants, sensitive equipment, or critical structural interfaces.

Under plate loading, the load introduction is more uniform and the crushing initiates across a wider region, which helps mitigate sharp peaks. In dynamic impact scenarios, by comparison, localised initiation, rate sensitivity, and inertia effects can elevate peak loads. Therefore, the lower PCF in the plate crushing case reinforces the idea that this loading condition promotes a gentler and more controlled onset of collapse.

Crushing Force Efficiency (CFE)

The Crushing Force Efficiency (CFE) of 0.363 indicates moderate crushing efficiency, meaning the ratio of mean force to peak force reflects a response that is reasonably stable but still leaves room for improvement. A moderate CFE suggests that while the crushing force is fairly well distributed during deformation, there is still a noticeable difference between initial peak behaviour and the subsequent mean crushing level. Nevertheless, the value supports the observation of controlled and progressive deformation, particularly when compared with scenarios that produce pronounced force spikes or unstable collapse patterns.

While CFE under axial plate crushing may be lower than under axial impact conditions, it still points towards a structurally sound performance: the panel maintains a balance between load-carrying capability and energy absorption, rather than prioritising one at the expense of the other. This balance is particularly useful in applications where both structural support and energy dissipation are required simultaneously.

Table 8.4: Plate Crush Analysis Result Values

Parameters	Values
Energy Absorption (J)	15,788.2
Specific Energy Absorption (J/kg)	25,337.74
Peak Crushing Force (N)	229,314
Mean Crushing Force (N)	83,219.7
Crushing Force Efficiency (CFE)	0.363

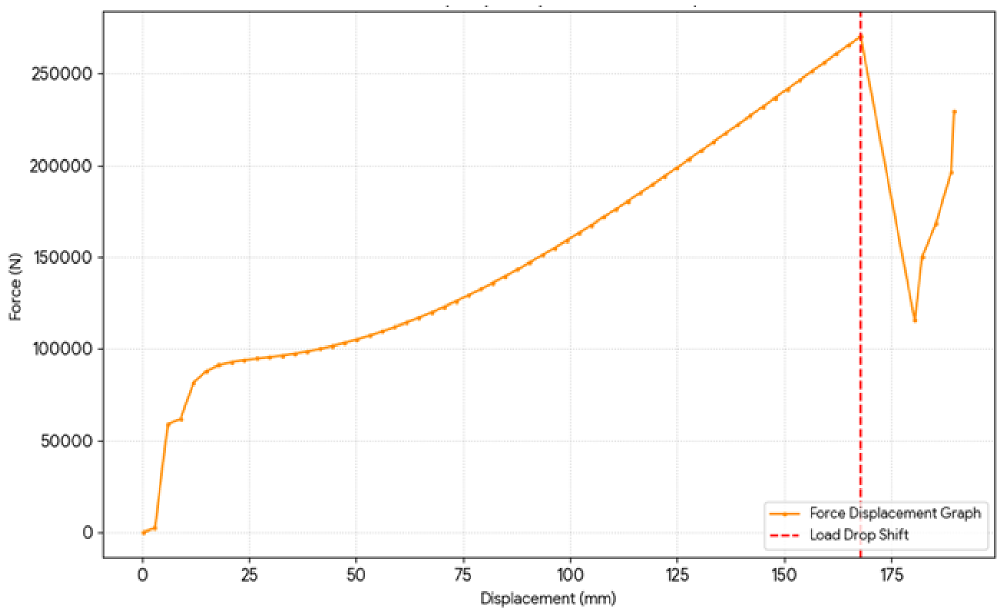

Figure 8.4: Force-Displacement Curve for the plate crush analysis

The force–displacement curve obtained from the axial plate crushing test, sown in fig 8.4, illustrates the deformation behavior and energy absorption characteristics of the hybrid foam-filled Voronoi sandwich panel.

Initial Region (0 – ~10 mm)

At the beginning of loading, the force increases rapidly from near zero to approximately 60,000 N. This region represents the initial contact and elastic deformation phase, where the structure begins to resist the applied load.

Plateau Region (~10 – 50 mm)

Following the initial rise, the curve shows a relatively gradual increase in force, stabilizing around 90,000–100,000 N. This indicates the onset of progressive collapse of the Voronoi cells, where the structure absorbs energy efficiently with relatively small fluctuations in force.

Progressive Hardening Region (~50 – 160 mm)

Beyond this stage, the force increases steadily with displacement, reaching a peak of approximately 270,000 N. This behavior corresponds to densification of the foam core and compaction of the cellular structure, where resistance increases as the material becomes more compact.

Peak Force and Drop (~160 mm)

At around 160 mm displacement, the structure reaches its peak crushing force (PCF). Immediately after this point, a sharp drop in force is observed, indicating structural instability or localized failure, such as buckling or fracture of the cell walls.

Post-Peak Region (>160 mm)

After the sudden drop, the force fluctuates and partially recovers, showing irregular behavior. This region represents continued crushing with damaged structure, where load-bearing capacity is reduced but still present.

The relatively smooth rise in force before the peak indicates good load distribution and controlled deformation, while the moderate peak force suggests acceptable impact performance.

Compared to dynamic impact conditions, this quasi-static response shows more stable and predictable deformation, which is desirable for applications requiring controlled energy absorption.

Overall Discussion

Overall, the axial plate crush analysis demonstrates that the hybrid foam-filled Voronoi sandwich structure performs effectively under compressive loading, with stable deformation and consistent energy absorption. The collapse mechanism is progressive, enabling gradual energy dissipation without severe force oscillations, and the foam-filled cellular design supports load redistribution that reduces the likelihood of catastrophic, localised failure.

When compared with axial drop ball impact, the plate crushing response is notably smoother and more controlled. It tends to produce lower peak forces, an advantage for limiting sudden load transmission, while showing slightly reduced energy absorption efficiency in specific terms. In summary, this hybrid structure is well-suited to applications that demand controlled deformation under compressive loads, such as engineered energy absorbers, protective panels etc.

CHAPTER 9

CONCLUSION AND FUTURE WORKS

This study investigated the crashworthiness performance of a hybrid foam-filled Voronoi sandwich panel under different loading conditions using numerical analysis. The structure was designed using bio-inspired Voronoi geometry and optimized through the Taguchi method to enhance its energy absorption characteristics.

From the results, it is observed that the structure exhibits significantly better performance under axial loading conditions. The axial drop ball impact analysis showed high energy absorption and a superior Crushing Force Efficiency (CFE ≈ 0.58), indicating stable progressive deformation and efficient load distribution. In contrast, surface impact resulted in very low CFE, reflecting poor energy absorption and unstable deformation behavior.

The axial plate crushing analysis further demonstrated that the structure undergoes controlled and progressive collapse with moderate CFE (≈ 0.36). The force–displacement behavior confirmed stable deformation with gradual energy dissipation and acceptable peak force levels. Compared to dynamic impact loading, the plate crushing response was smoother and more predictable, highlighting the influence of loading conditions on structural behavior. Overall, the hybrid Voronoi–foam sandwich structure offers:

- Improved energy absorption
- Reduced peak crushing force
- Stable progressive deformation

These characteristics make it a promising candidate for crashworthy applications such as automotive crumple zones and impact protection systems.

Although the present study delivers valuable insights into the proposed structure, there is clear scope for further refinement and deeper investigation to strengthen both performance and real-world applicability. Several targeted avenues of future work could help validate the findings, reduce uncertainty, and broaden the structure's potential use cases across demanding engineering environments.

- **Experimental Validation**

  Carry out controlled physical testing to corroborate the numerical simulation outcomes. Comparing test data with model predictions would support calibration of material models, boundary conditions, and failure criteria, ultimately improving overall accuracy and confidence.

- **Material Optimisation**

  Examine alternative material systems, including advanced composites, hybrid laminates, foam density or functionally graded materials. This could increase specific energy absorption while reducing mass, and may also improve corrosion resistance or durability depending on the application.

- **Geometric Optimisation**

  Systematically explore different Voronoi morphologies, cell densities, strut arrangements, and thickness ratios. Parametric studies and optimisation algorithms could be used to identify configurations that best balance strength, stiffness, crush efficiency, and manufacturability.

- **Multi-Impact Analysis**

  Assess structural response under repeated impacts, off-axis loading, or multi-directional collision scenarios. This would help determine whether the structure maintains performance after initial damage and how failure modes evolve over successive events.

- **Manufacturing Feasibility**

  Investigate practical fabrication routes, particularly additive manufacturing methods suited to complex lattices. This include tolerances, repeatability, post-processing, cost, and scalability for production.

CHAPTER 10

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PROGRAM OUTCOMES (POs)

PO1 – Engineering Knowledge

Core mechanical concepts like crashworthiness, material behavior, and energy absorption are applied. Fundamental engineering principles are used in design and analysis.

PO2 – Problem Analysis

The project involves identifying crash performance issues and analyzing them using simulation and optimization techniques. Engineering problems are solved using analytical approaches.

PO3 – Design/Development of Solutions

Voronoi foam sandwich structures are designed to improve energy absorption. The design meets safety and functional requirements under impact conditions.

PO4 – Conduct Investigations

Simulation studies and result evaluations are carried out to investigate crash performance. Data interpretation helps in validating the design effectiveness.

PO5 – Modern Tool Usage

Software tools like ANSYS/Abaqus are used for modeling and simulation. These tools help in predicting structural behavior under crash conditions.

PO6 – Engineer and Society

The project contributes to vehicle safety by improving crashworthiness. It addresses societal concerns related to accident impact and passenger protection.

PO7 – Environment and Sustainability

Lightweight structures reduce material usage and improve fuel efficiency. This supports environmentally sustainable engineering solutions.

PO8 – Ethics

Engineering design is carried out considering safety and professional responsibility. Ethical practices are followed in simulation and reporting.

PO9 – Individual and Team Work

The project is executed through teamwork with proper task distribution. Collaboration ensures efficient completion of different project stages.

PO10 – Communication

Project findings are documented and presented clearly through reports and presentations. Effective communication of technical results is ensured.

PO11 – Project Management and Finance

Project planning, scheduling, and resource management are followed. Basic management principles are applied to complete the work on time.

PO12 – Life-long Learning

New tools, simulation methods, and optimization techniques are learned during the project. This enhances continuous learning ability.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1 – Core Mechanical Engineering Knowledge

The project applies concepts from design, materials, and mechanics to solve engineering problems. It strengthens core mechanical engineering fundamentals.

PSO2 – Modern Engineering Tools

Advanced software like ANSYS/Abaqus is used for analysis and validation. The project enhances skills in modern design and simulation tools.

PSO3 – Societal and Environmental Application

The design improves safety and promotes lightweight structures for real-world applications. It addresses societal needs and environmental considerations.

COURSE OUTCOMES(COs)

At the end of the course, I am able to understand the following outcomes:

Course Outcome	**Description**
CO1	Develop a clear understanding of crashworthiness principles and key energy absorption concepts.
CO2	Learn how to design and generate Voronoi-based cellular structures for lightweight protection.
CO3	Apply optimisation techniques, including the Taguchi method, to refine designs efficiently.
CO4	Carry out numerical simulations using ANSYS & Abaqus
CO5	Evaluate performance metrics and confidently interpret the results to guide improvements

MAPPING COURSE OUTCOMES WITH PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES

CO/PO &PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	1	1	-	-	1	-	2	3	2	1
CO2	2	3	3	2	2	1	1	-	1	1	-	2	3	3	1
CO3	2	3	2	3	2	1	-	1	2	2	1	3	2	3	1
CO4	2	2	2	3	3	1	-	1	2	1	2	2	2	3	2
CO5	3	3	3	2	2	2	1	1	2	2	1	3	3	2	3
CO avg	2.4	2.6	2.5	2.5	2.25	1.2	1	1	1.75	1.4	1.3	2.4	2.6	2.6	1.6

SUSTAINABLE DEVELOPMENT GOALS (SDG) ALIGNMENT OF THE PROJECT

Project Title: Design and Impact Analysis of Hybrid Foam-Filled Voronoi Sandwich Panels for Vehicle Crumple Zones

SDG No	SDG Title	Relevance to the Project
SDG 9	Industry, Innovation and Infrastructure	The development of advanced lightweight energy-absorbing structures improves vehicle safety, promotes innovative engineering design, and supports sustainable industrial development.

Justification for Project Relevance:

- The project introduces **innovative bio-inspired Voronoi structures**, contributing to advancements in modern engineering design.
- The use of **lightweight sandwich panels** enhances fuel efficiency, reducing overall energy consumption in vehicles.
- Improved **crashworthiness and energy absorption** increases passenger safety, supporting resilient infrastructure in transportation systems.
- The integration of **foam-filled structures** optimizes material usage and structural efficiency, promoting sustainable manufacturing practices.
- The project supports **technological innovation in the automotive sector**, aligning with the goals of smart and sustainable industrial development.

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Department of Mechanical Engineering

SRM Nagar, Kattankulathur - 603203, Tamil Nadu, India

6TH INTERNATIONAL CONFERENCE ON ADVANCES IN MECHANICAL ENGINEERING (ICAME-2026)

Certificate of Presentation

This is to certify that Prof./Dr./Mr./Ms./Mrs **SRIKANTH R**

has presented a paper (ID: ICAME-M154) titled **Design and Impact Analysis of Hybrid Foam-Filled Voronoi Sandwich Panels for Vehicle Crumple Zones**

in the 6th **International Conference on Advances in Mechanical Engineering (ICAME-2026)**, organized by the Department of Mechanical Engineering, SRM Institute of Science and Technology, Kattankulathur, Tamil Nadu, India, in association with the University of Split, Croatia, during March 18–20, 2026.

Dr. K. Suresh Kumar
Convener, ICAME 2026
Professor & Head
Department of Mechanical Engineering

Dr. M. Cheralathan
Professor & Chairperson
School of Mechanical Engineering

NAAC	UGC	NIRF	QS	THE World University Rankings	Shanghai Ranking	Nature Index	UI GreenMetric
A++	Category I with 12B Status	(2025) Ranked 11th University	(2026) World Ranking one among 44 Indian Universities	(2025) World Ranking one among 91 Indian Universities	(2025) World Ranking Ranked 8-9 in Indian Universities	(2025) World Ranking one among 18 Indian Universities	(2024) World Ranking one among 3 Indian Universities

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This is to certify that Prof./Dr./Mr./Ms./Mrs **YUVAN SHANKAR T**
has presented a paper (ID: ICAME-M154) titled **Design and Impact Analysis of Hybrid Foam-Filled Voronoi Sandwich Panels for Vehicle Crumple Zones**
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